ExoInstruments

Player Guide

Real telescopes, from a backyard astrograph to orbit, for Kerbal Space Program.

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1 Overview

ExoInstruments adds a working observatory to Kerbal Space Program.

What it adds has two halves. The first is **observational astronomy**: a roster of named telescopes, from a backyard astrograph to the Very Large Telescope to a space telescope you launch yourself, that take genuine photographs of anything in the sky: planets, moons, star fields, nebulae, galaxies. The light goes through each instrument's own optics onto its detector model, and the frames come out as FITS files that real astronomy software opens, calibrates and stacks. The second is an **exoplanet survey**: select a star from a published catalog, buy time on an instrument suited to its brightness, and extract the signal from simulated, photon-noise-limited data the way an observational astronomer would. Every number in both halves comes from the instruments' own published papers.

Everything runs from the **Observatory**, a dedicated facility at the space center. Click the building itself, or the mod's button on the toolbar, while in the Space Center scene.

Game modes. In **Career**, star identities are hidden until observed (see section 3), instruments must be purchased, observations cost Funds, and discoveries pay Science. In **Science** mode every instrument is available and observations are free, but discoveries still pay Science. In **Sandbox** everything is available and nothing is charged or awarded. The Science and Funds gain sliders of your difficulty settings apply to all rewards.

A first detection, in six steps

1. Open the Observatory. In career you start with WASP, a wide-field transit survey camera whose observations cost almost nothing.
2. In the target search, type `mag:<8 alt:>40` and pick a star: bright, and high in the sky right now. In career it will only show a provisional designation; that is the fog of war, and lifting it is the job.
3. Press *Start Observation*, then warp. Points accumulate at night while the target is up; the forecast heatmap's best-window button takes you straight to good conditions.
4. Give it several game days. A transit search needs the planet to cross more than once, so a baseline of two or three orbital periods is the difference between a detection and noise.

5. Press *Stop Observation*, then *Run Detection Analysis*.
6. Read the verdict. A signal-to-noise ratio of 8 or better on a catalog planet is a confirmed detection and pays the bonus; either way the star is now identified, paid, and permanently revealed on your chart.

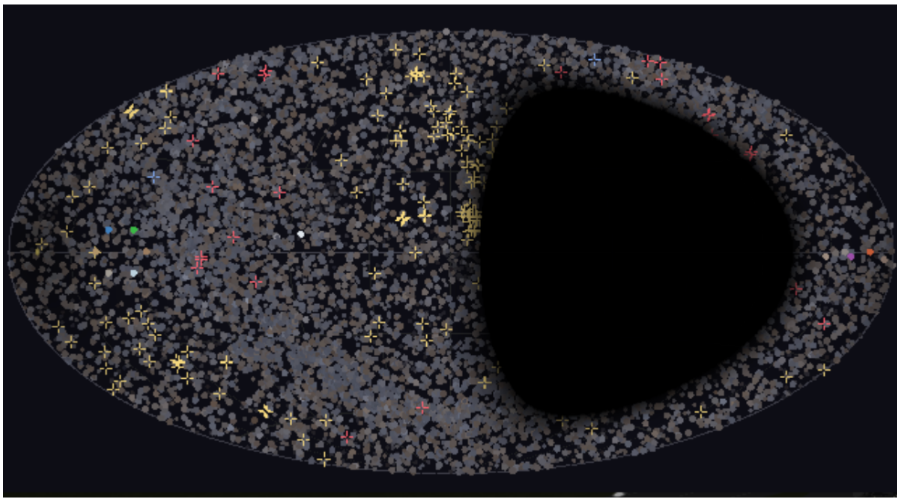
Most stars in the catalog host no known planet, and a null result on one still pays the survey award: ruling stars out is how a survey narrows the sky.

2 The Observatory window

Instrument selector. The header row picks the active instrument. In career, locked instruments appear as rows stating their price and the Science track record required; the Unlock button activates when you can afford both.

Sky chart. The left panel is a chart of everything the active instrument can point at: catalog stars, and for the photographic instruments also the planets and moons of your solar system, galaxies and nebulae. Click a marker to select it.

Target search. The right panel searches everything by designation (`M31`, `NGC 224` and `Andromeda` find the same entry), by type (`type:galaxy`), constellation (`in:Ori`), magnitude (`mag:<9`) or current altitude (`alt:>30`). Clicking a result points the telescope at it.



The full-sky chart: catalog stars, solar-system bodies and deep-sky objects on one map. The dark region is the sky currently blocked by the planet itself.

Observing forecast. For a selected target, the forecast heatmap shows observing quality for the nights ahead: rows are nights, columns are time of night, folding in twilight, target altitude, airmass and lunar interference. Click any cell, or the best-window button, to warp directly there. If BetterTimeWarpContinued is installed, all warps in the mod use its higher rates automatically.

Ground-based constraints. The detection instruments collect data only while the Sun is below nautical twilight ($-12°$) and the target is above 20° altitude. Expect diurnal gaps in your light curves; they are real, and so are the aliases they create in period searches. Both of Kerbin's moons occult targets outright and brighten the sky around themselves; a full Mün near your target measurably degrades photometric noise.

3 Career progression

Fog of war. In career, an unobserved star shows only a provisional designation built from its position (for example "Unscanned J2257+2046"). Its name, catalog status and system contents are revealed by completing an observation and its analysis, whatever the outcome. A large catalog of ordinary background stars is blended in, so a revealed planet host is a genuine discovery, not a matter of clicking every marker.

What things cost. Each instrument has a one-time acquisition price plus a cumulative Science threshold, and each observation bills telescope time. Prices follow the published scaling law relating telescope cost to aperture (van Belle, Meinel and Meinel 2004), compressed to fit a KSP career, so the ladder climbs in the order real facilities cost.

Instrument	Method	Availability	Unlock	Science req.	Per obs.
WASP	Transit	from start	0	0	20
RedCat 51	Photography	from start	0	0	20
RC20	Photography	purchase	15 000	10	20
CDK1000	Photography	purchase	44 000	25	45
TESS	Transit	purchase	96 000	50	950
SOPHIE	Radial velocity	purchase	126 000	60	840
SPECULOOS	Transit	purchase	133 000	65	1 330
HARPS	Radial velocity	purchase	340 000	120	2 275
VLT FORS2	Photography	purchase	1 270 000	250	1 270
VLT SPHERE	Photography	purchase	1 590 000	300	1 590
ESPRESSO	Radial velocity	purchase	1 910 000	350	12 700
ELT	Direct imaging	under construction	not yet offered		
Orbital Observatory	Photography	launch it yourself	see section 7		

The Science threshold counts Science earned through this program, not your R&D balance, so spending points in the tech tree never locks you out of an instrument you have already earned.

What discoveries pay. The first completed scan of any star pays a survey award (full value for roughly your first fifty stars, then diminishing; ruling a star out is survey work too, but the thousandth null result is not news). A confirmed planet detection pays a one-time bonus per host, scaled by the instrument's own multiplier, and it is checked against the catalog: a statistical false positive on an empty star pays nothing. Resolving several planets of one system in a single campaign pays a jackpot on top. Transit timing variations and Rossiter-McLaughlin measurements (below) each carry their own one-time award per system.

4 Astrophotography

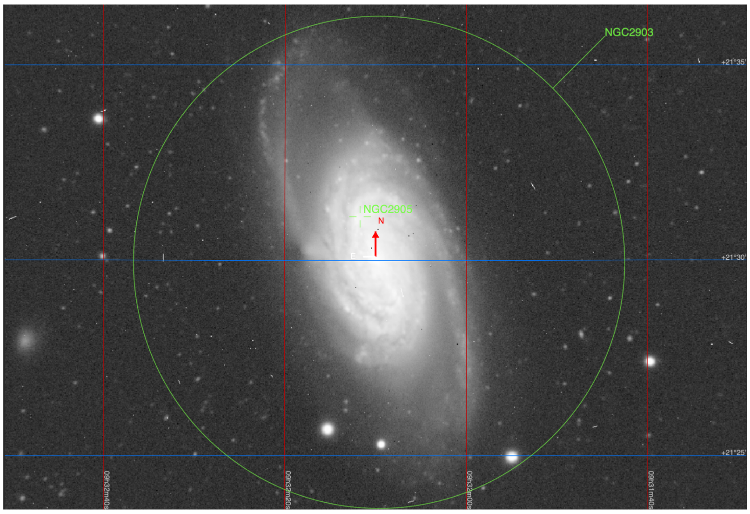
The photographic instruments take a live, physically honest photograph of anything you point them at: a planet or moon at its true position, phase and brightness, or a deep-sky field, star clusters, nebulae, galaxies, drawn from survey catalogs. The light goes through the telescope's optical train onto its sensor model, and what comes out is what a camera on that telescope would actually record.

Taking a frame. Select a photographic instrument, click a target on the chart (or search for it), and open the camera panel. You control exposure time, gain, the filter wheel, focus, and on instruments that have one, pixel binning and a Barlow or focal-reducer stage. The exposure then runs for the time you set: nothing renders early. The target must be above the horizon, and the sky model includes twilight, airglow, zodiacal light and moonlight, so a frame taken into a bright sky washes out exactly as it should.

A bright star through the RC20. The four diffraction spikes are the spider vanes of the instrument's own pupil; the grey background is the display's asinh stretch lifting the sky's true level, not an error.

The frame is honest. Shot noise, dark current, read noise, hot pixels, cosmic rays, blooming, atmospheric seeing (or an adaptive-optics correction, on SPHERE), and star trailing when you shoot unguided: all are modeled on the published detector literature. What you get is one monochrome, grainy, unprocessed sub, saved as PNG and as a 16-bit FITS file with standard header keywords and world coordinates, the same product a real camera writes. Everything lands under `Screenshots/` in your KSP install; stacking sessions get their own subfolder.

The FITS files are real enough for real software: they open, calibrate and stack in the same tools actual astrophotographers use, such as Siril, and the world coordinates in the header let those tools plate-identify the field on their own.

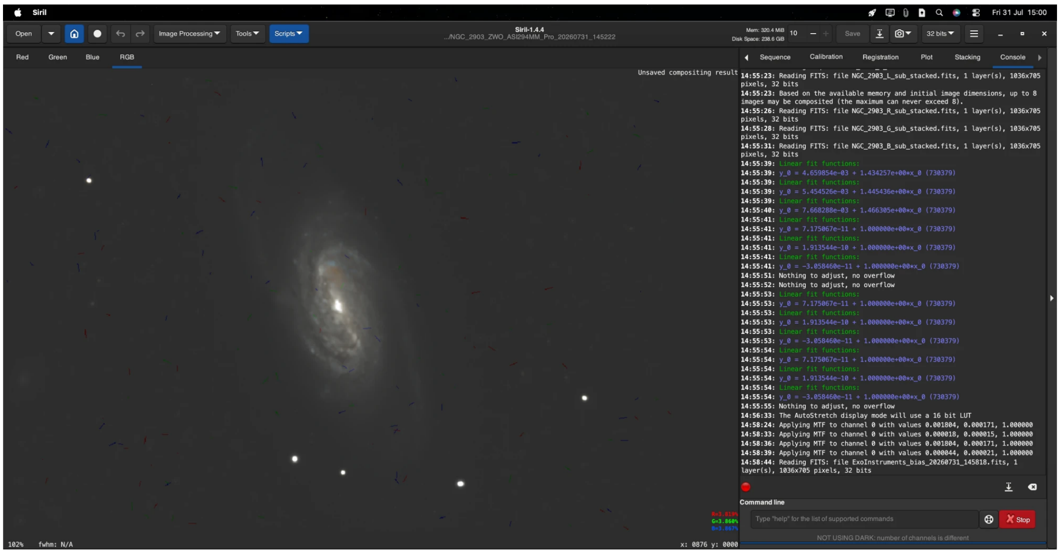


A single luminance sub of NGC 2903, identified and annotated by external astronomy software entirely from the world coordinates the mod wrote into the FITS header.

Brightness management. Kerbin's moons sit far closer to their star than any real moon and will saturate a naive exposure. Watch the live saturation readout and use the neutral-density filter wheel; the ND options run to a solar-grade ND 5.0.

Choosing the instrument. Framing beats aperture. The RC20 and CDK1000 frame large, nearby targets (Minmus wide, Eve, Jool, Duna tight); FORS2 adds the light grasp for small and distant bodies; SPHERE's adaptive optics resolve the dwarf-planet class (Dres, Eeloo, Gilly, Jool's small moons) that nothing else can, in a field only 3.6″ wide. The README carries per-target tables for both the stock system and RSS.

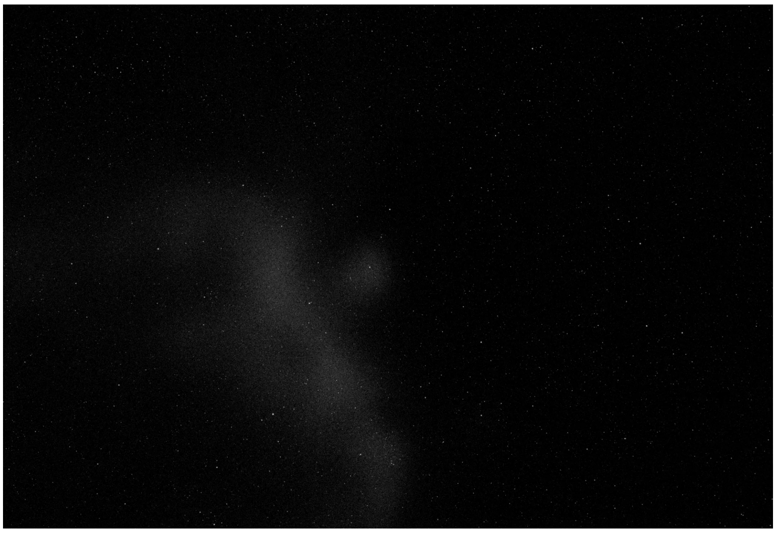
Stacking. The RC20 panel can capture a series of subs per filter and combine them: bad-pixel correction, optional centroid alignment, background subtraction, and LRGB composition with an optional Hα blend into red, plus an optional lucky-imaging mode that keeps only the sharpest subs before stacking. Many short exposures beat one long one, for the same reasons they do in real astrophotography.



The same NGC 2903 session finished in Siril: the mod's own bias, dark and flat frames calibrating the subs, the four filters stacked, and the channels composed into RGB. The whole pipeline runs on the mod's FITS output exactly as it would on a night's data.

Supernovae. With the galaxy catalog installed, its galaxies host supernovae at their own real rates, and nothing tells you where one is. You find them the way they are really found: by photographing galaxies and looking at what came back. A point of light on a galaxy that the frame detects at five sigma, against the host's own glow at that spot, is logged as a discovery, pays Science once, and is then marked on the chart while it lasts. Each save has its own fixed history, so reloading never rerolls it, and a galaxy the catalog gives no distance for hosts none at all.

The sky behind the subject. Star fields, the dust map, the Hα emission map and the galaxy catalog are optional data products built from published surveys. Nothing ships in the download; one script (`tools/setup_data.py`, see the README) builds them. Without them the mod works fully and the photographic sky is simply empty behind the target.



What the optional data buys: the Seagull Nebula's emission through the RedCat 51 wide-field astrograph, over a Gaia star field.

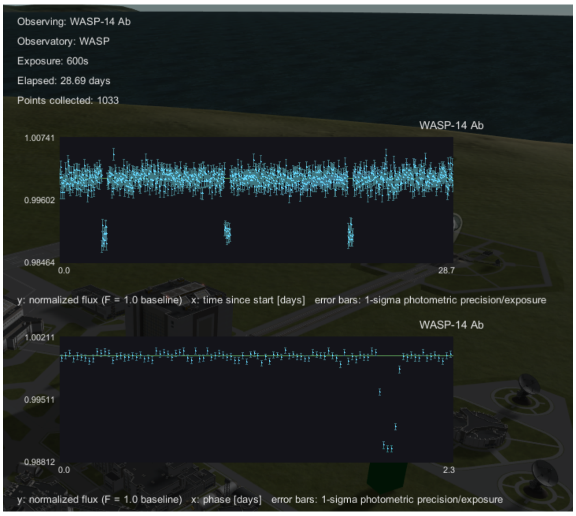
5 Transit photometry

The transit method watches a star's brightness for the periodic dip of a planet crossing its disk. It is the workhorse of the program and the right place to start.

Running a campaign.

1. Select a transit instrument and click a star on the chart.
2. Press *Start Observation* and engage time warp. Points accumulate while the target is observable; a useful baseline is many days, since you need several transits, not one.
3. Press *Stop Observation*, then *Run Detection Analysis*.

The analysis runs a Box Least Squares period search over your actual samples. A detection needs a signal-to-noise ratio of 8, so a short or badly scheduled campaign genuinely fails: too few points, a baseline shorter than two periods, or noise from moonlight can all bury a real planet. The report shows the raw light curve, the phase-folded curve at the best period, and depth, period and SNR. Elapsed time is shown in your game's own day length (six-hour days on the default Kerbin clock).



A transit campaign: raw time series above, the phase-folded transit below.

Observing from orbit. TESS is the one transit instrument in space, and that is what its price buys: it bypasses the twilight and altitude gates entirely and collects around the clock, so its light curves have none of the diurnal gaps, and none of the window-function aliases, that complicate every ground-based period search.

What you are fighting. Photon noise set by the star's magnitude and the instrument's aperture; atmospheric scintillation on ground instruments; moonlight; and the star itself, which carries persistent spot modulation. The detector rejects the classic false positive, a starspot masquerading as a transit, through a duty-cycle test, so not every dip survives analysis.

Multi-planet systems. Compact systems superpose every transiting member on one light curve. The analysis detects the strongest signal, masks its transits, and re-searches the residual, up to four planets. Confirming several at once pays the jackpot bonus.

Transit timing variations. With six or more measured transits of one planet, the analysis re-fits the ephemeris and searches the timing residuals for a periodic wobble: the gravitational signature of another body pulling the transiter around. A confirmed TTV signal is evidence of a companion you may never see transit, and pays its own award.

6 Radial velocity

A spectrograph does not watch brightness; it measures the star's line-of-sight velocity as the planet's gravity swings it around the common center of mass. It needs no transit geometry, so it reaches planets the transit method is blind to, and it yields the planet's minimum mass.

The campaign flow is the same: point, observe across many nights, stop, analyze. The detector fits sinusoids across a period grid, with guards against the phantom periods that sampling cadence creates. Multi-planet systems are resolved by prewhitening: the strongest signal is fitted and subtracted, and the residuals are searched again, up to four planets. The report flags any candidate whose period is a suspicious integer ratio of another; deciding whether that is a resonance or a harmonic is your call, as it is in real surveys.

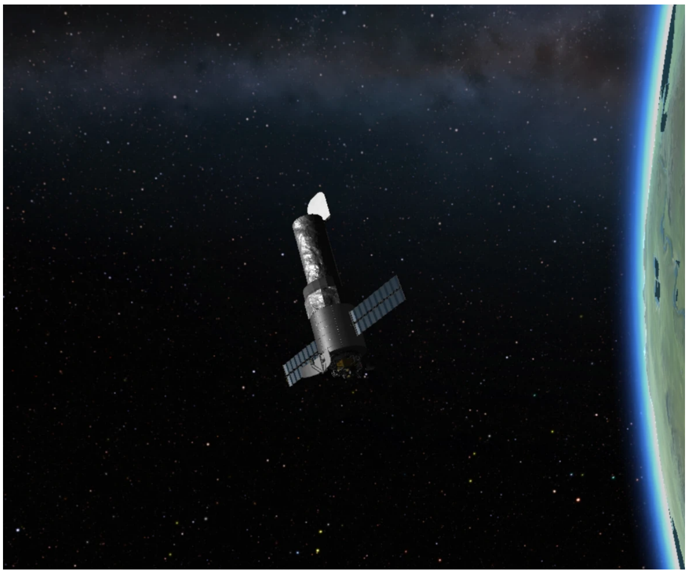
The noise floor is the star itself as much as the instrument: every star carries persistent activity jitter, so a quiet star at 8th magnitude is a better target than an active one at 7th. The instrument ladder (SOPHIE at roughly 2 m/s, HARPS at 1 m/s, ESPRESSO at 0.15 m/s) is what determines how small a planet you can pull out of that scatter.

Rossiter-McLaughlin. When a transiting planet crosses its rotating star, it blocks first the approaching and then the receding limb, imprinting a characteristic anomaly on the RV curve during the transit. Measuring it gives the spin-orbit angle: whether the planet orbits in the plane its star rotates in. This is the program's cross-method payoff: it requires a transit ephemeris you have already measured photometrically, and a spectrograph on the target while a transit is in progress; the mod schedules the burst sampling from your own ephemeris. A confirmed measurement carries its own award.

7 The orbital telescope

Every other instrument is somebody else's telescope whose time you buy. This one does not exist until you launch it.

The part. *Orbital Astrophysics Observatory*, in the Science category, unlocked at the Advanced Science Tech node. It is a 2.4 m Ritchey-Chrétien optical tube assembly carrying a wide-field UV/visible imager, its aperture door, and its own reaction wheels; 4.5 t, 3.75 m bulkhead, with a single bottom attachment node. It appears in the Observatory's instrument list only once one is in orbit with its door open. There is no unlock fee: the cost is the part, the launcher, and building a spacecraft that holds a target still.



An Orbital Astrophysics Observatory on station: door open, panels and antenna clear of the tube, wheels holding the target.

Four requirements, checked and reported by name when they fail:

1. **Aperture door open.** A hard gate, toggled in flight or by action group.
2. **A clear view down the tube.** Rays are cast across the pupil into your actual vessel. A solar panel or antenna mounted over the open end blocks the telescope, and the panel names the offending part. There is no partial credit: an obstructed pupil is a different telescope, not a dimmer one.
3. **Attitude control that can hold a target.** The wheels hold the boresight to HST's published $0.008''$ jitter, a fifth of a pixel. Thrusters cannot hold a point at all: an on-off thruster drifts across its deadband and is pulsed back forever, which at this plate scale is hundreds of pixels of smear. Keep the wheels; add more for a heavy bus.
4. **Electric charge.**

Operating it. Flying the spacecraft yourself, power and a clear aperture are enough. From the Observatory on the ground, every command and every frame goes over radio, so the vessel also needs an antenna with a working CommNet link. A frame is 32 MB: about nine minutes through a Communotron 16, about two through a relay dish. The antenna you chose is a real decision.

Repointing costs time and power. A slew runs at HST's published rate limit, scaled to your home world so it occupies the same fraction of an orbit as it does for the real spacecraft, and every repoint pays a fixed guide-star acquisition on arrival, which is why observing programs cluster nearby targets instead of touring the sky. The wheels bill the battery for the whole manoeuvre; a large repoint costs a substantial fraction of the part's own charge, and the panel refuses a slew the spacecraft cannot finish. Solar panels recharge the vessel between commands, on an orbit-averaged ledger that accounts for eclipse.

Sky constraints no ground telescope has. The planet occults most targets for part of every orbit. You cannot point within 62.5° of the Sun, 20° of the sunlit limb, 7.6° of the dark limb, or 9° of a moon. The observing window is finite and the panel shows how long the current orbit lets you integrate; a target near your orbital pole lies in the continuous viewing zone and is never occulted at all.

What it buys you. Not resolution: at 2.4 m it does not out-resolve the VLT. What it has is the near-ultraviolet down to 200 nm, which the atmosphere blocks outright at any altitude; a point-spread function identical in every frame ever taken, because there is no seeing; and a sky about 1.6 magnitudes darker, because airglow is something an atmosphere does. Exposures taken while slewing are not forbidden; they come out streaked, as they would.

8 Reading the reports

Every analysis states its statistics plainly: SNR against the detection threshold, number of samples, baseline, and for transits the phase-folded curve your eye can judge. Two habits carry over from real practice. First, a non-detection is information; the star is surveyed, revealed and paid either way. Second, more baseline beats more instruments: most failed campaigns are short campaigns.

9 Troubleshooting

No points accumulate during a campaign. It is daytime, or the target is below 20° altitude. Open the forecast heatmap and warp to the next good window. Space-based instruments do not have this constraint.

The analysis finds nothing on a star I know has a planet. Almost always baseline: the search needs several complete orbital periods, not several nights. Extend the same campaign rather than restarting short ones, and avoid nights with a bright moon near the target.

The photograph is pure white. Saturated. Shorten the exposure, reduce gain, or select a neutral-density filter; Kerbin's own moons need aggressive ND by design.

The photograph is black. The exposure is too short for the target's brightness, or the target sits below the horizon. Check the live signal and saturation readouts before committing a long exposure.

Stars and the target trail across the frame. You are unguided on a long exposure. Enable the autoguider where the instrument has one, or take shorter subs and stack them.

The orbital telescope does not appear in the instrument list. It must be in orbit with its aperture door open. If it still refuses, the panel names the cause: an obstructing part over the pupil, no attitude control, or no charge.

I cannot command the orbital telescope from the Observatory. No working CommNet link. Add an antenna, point it home, or fly the spacecraft and operate it directly.

The sky behind my photographs is empty. The optional star catalog is not installed. Nothing ships with the mod; run `tools/setup_data.py` (see the README) to build it, or keep playing without it, since nothing else depends on it.

Questions, bug reports, ideas: `@Gratiste` on Discord.

Fly safe.

-Baptiste